EXPERIMENT NO. 38
CONCAVE GRATING SPECTROMETER, ROWLAND MOUNTING

I Objectives

(i) To show that the variation of wavelength with respect to linear as well as angular position of the eyepiece is inversely proportional to the spectral order.

(ii) To plot a calibration curve for the grating spectrometer and to use it in the determination of the wavelengths of an unknown source.

II Apparatus

Concave grating spectrometer, mercury vapour lamp source, an "unknown" source, table lamp and condensing lens are required in the experiment.

The principal piece of apparatus is a Rowland Mounting spectrometer equipped with a concave reflection grating. By means of the screws at the back of the grating holder the grating may be rotated about two mutually perpendicular axes, one vertical and the other horizontal in the plane of the grating. The grating may be rotated about its optical axis by means of two screws at the top of the grating holder. See the apparatus yourself and find out the details. This adjustment is important since the lines of the grating must be perpendicular to the plane of the principal component parts, viz, slit, grating and eyepiece. The grating must have a radius of curvature equal to the diameter of the Rowland circle of the mounting. The grating has a focal length of 21.25 cm and 15,000 lines per inch. The track scale has a linear calibration. The circular scale for measuring angular separation $d\theta$ is graduated in degrees.

The slit has two adjustments. One knurled screw controls the width of the slit and the other knurled screw effects the rotation of the slit about the line joining the slit and the grating. This is quite important if sharp lines are to be obtained in the focal plane of the eyepiece. Any convenient line-spectrum source may be used but a mercury vapour lamp source is recommended for making the final adjustments.

III Adjustments

Place a small light source slightly above and behind the grating holder so as to illuminate the cross wires in the eyepiece. Looking into the eyepiece (which should be so adjusted as to put the cross wires in focus) adjust the screws at the back of the grating holder until the image of the cross wires themselves are in perfect coincidence. For this, it may be necessary to move the grating towards or away from the eyepiece. This can be done by releasing the screw (the one which is nearer to the grating holder) on the arm joining the eyepiece and the grating holder. There will be no parallax between the cross wires and their image if the adjustment is good. These adjustments bring the optical axis of the grating into coincidence with the intersection of the cross wires. The small light source is no more needed.

Place the mercury source and the condensing lens in front of the slit. Adjust the lens and the slit width to ensure that the whole grating is illuminated. Make sure no spurious light falls on the grating. Look for the diffraction spectrum by moving the eyepiece holder on the railing. You must be able to see various lines of the mercury spectrum and at least two orders (I and II). Coincide the 546.1 nm (green) line of the spectrum with the cross wire for

final adjustments. It may be necessary to orient the slit and the grating so that the slit, rulings of the grating and the vertical cross wires are parallel. A narrow slit will now produce a well defined sharp image of the slit at the cross wire in green colour. If the adjustments are perfect you will be able to see the two orders of spectrum by moving the eyepiece along the track appropriately. Now your grating spectrometer is ready for measurements.

IV Measurements and Analysis

A. Spectral study with the direction of observation fixed along a diameter of the Rowland circle: Measure the linear distance x of the spectral lines from the slit for the I-order and the II-order spectrum. Identify the wavelengths of the spectral lines with the help of Table I. Plot a graph of wavelength against the linear distance x from the slit for each spectral order, n, and show that (Ref .1)

$$(d\lambda/dx) \propto 1/n.$$

B. Spectral study with the angle of incidence constant and moving the eyepiece along an arc of the Rowland circle: Coincide the cross wires with one of the spectral lines in the middle of the Ist-order spectrum (either Indigo or Green). Now measure the angular position of the spectral lines with respect to the chosen line. Do the same for the IInd-order spectrum. Plot a graph of wavelength against the angular position θ, and show that

$$(d\lambda/d\theta) \propto 1/n.$$

C. Calibration curve for the grating spectrometer: Using the data for part A above, plot a full page calibration curve of λ against x, for the Ist and IInd order spectrum of mercury. Similarly calibrate the circular scale, i.e., λ vs. θ.

D. Determination of the unknown wavelength: Now replace the mercury lamp with the unknown source (here we assume the sodium vapour lamp as the unknown source). Record the linear positions of the spectral lines of the unknown source as in Part A above. With the help of the calibration curves (Part C) determine the wavelengths of the unknown source. Make use of the calibration curves for the I and II-order spectra.

Table I: The standard lines of mercury spectrum.

Wavelength (nm)	Colour	Relative Intensity
404.7	Violet [DEEP]	1800
407.8	Violet	150
435.8	Indigo [VIOLET]	4000
491.6	Blue-green [DEEP GREEN]	80
546.1	Green [YELLOWISH GREEN]	1100
577.0	Yellow-orange	240
579.1	Yellow-orange	280
623.4	Red	
690.8	Red	

CAUTION: WHILE MAKING THE ADJUSTMENTS AND TAKING READINGS SEE THAT YOU DO NOT TOUCH OR IN ANY WAY DAMAGE THE RULED SURFACE OF THE GRATING. JUST LEAVE IT ALONE EVEN IF YOU SEE DUST ON IT.

V Questions

1. What are the advantages and disadvantages of the concave grating over (i) a plane grating (ii) a prism?
2. What are the values of angular dispersion and resolving power of the grating used in the region of the first order green line of mercury?
3. How would the spectrum be modified if each of the following changes are made independently?
   (a)The area of the grating is doubled.
   (b)The number of lines on the grating is doubled.
   (c)The separation between adjacent lines of the grating is doubled.
   (d) The separation between adjacent lines of the grating is halved.
4. What would be the minimum width of the grating to just resolve the yellow lines of (i) mercury (ii) sodium?
5. Name some of the mountings used for concave grating and mention why the mounting used in the present set-up is preferred over the others.

VI References

1. F. A. Jenkins & H. E. White, "Fundamentals of Optics," Ch. 17, pp 349.
2. M. Born and E. Wolf, "Principles of Optics," Ch. 8, pp. 418.

Useful Data: Grating element: 15000 lines/inch
Focal length : 21.25 cm
Length of grating: 1.8 cm